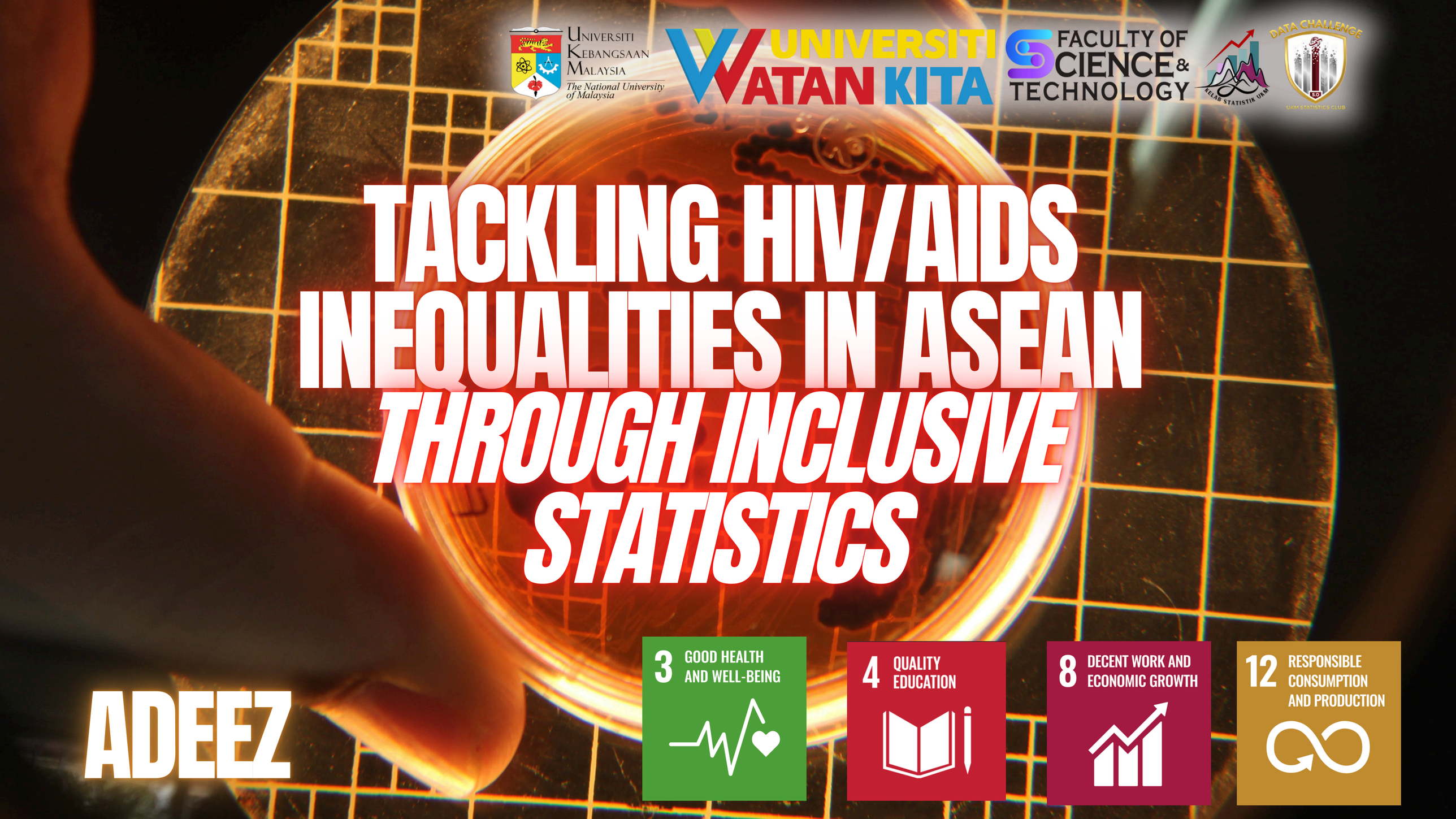
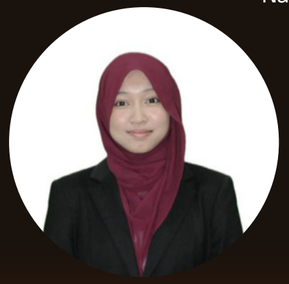

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia
UNIVERSITI WATAN KITA
FACULTY OF SCIENCE & TECHNOLOGY
KELAB STATISTIK UKM
DATA CHALLENGE
UKM STATISTICS CLUB
TACKLING HIV/AIDS INEQUALITIES IN ASEAN THROUGH INCLUSIVE STATISTICS
ADEEZ
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
8 DECENT WORK AND ECONOMIC GROWTH
12 RESPONSIBLE CONSUMPTION AND PRODUCTION

ADEEZ TEAM MEMBERS

DR WAN NURAINI FARHANA BINTI WAN NASIR

Naufal Akasyah bin Bakri

Mohamad Afiq Muttaqin Bin Muttacohiri



Nurul Dyaasykin Binti Mohd Ezany

Nur Ayuni Binti Azizi

Che Wan Nur Dayana Binti C Wan Roslan

OBJECTIVES OF THE STUDY CASE

ANALYZE

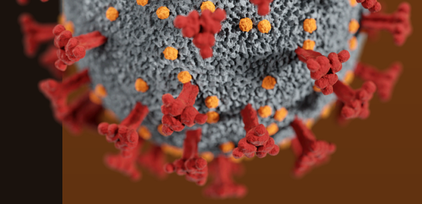
To examine the distribution of HIV/AIDS cases and treatment across ASEAN countries using statistical data.

DATA-DRIVEN STRATEGIES

To propose data-driven recommendations to reduce HIV/AIDS inequalities in ASEAN based on inclusive statistical analysis.

EVALUATION

To evaluate how inclusive statistics can help highlight health disparities related to HIV/AIDS in the ASEAN region.



AIDS

(ACQUIRED IMMUNODEFICIENCY SYNDROME)

When HIV is not treated, it slowly damages the immune system until the body can no longer fight infections. At this stage, the person develops AIDS.

People with AIDS may develop opportunistic infections, which are diseases that attack when the immune system is very weak, such as:

- Tuberculosis (TB)
- Pneumonia
- Certain cancers
- Severe infections

The Star Young adults majority of new HIV cases in Malaysia

Young adults majority of new HIV cases in Malaysia

By KHOO GEK SAN, ARFA YUNUS and JUNAID IBRAHIM
Thursday, 07 Aug 2025 | 9:11 AM MYT

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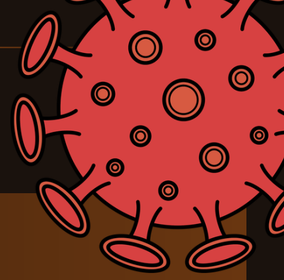


HIV

(HUMAN IMMUNODEFICIENCY VIRUS)

A virus that attacks the immune system, especially the CD4 cells (T-helper cells) that help the body fight infections.

Common ways HIV spreads:

- Unprotected sexual contact with an infected person
- Sharing needles or syringes
- From mother to baby during pregnancy, birth, or breastfeeding
- Blood transfusion with infected blood (rare today because blood is screened)

SOURCE: WORLD HEALTH ORGANISATION (WHO)

A person can live with HIV for many years if they receive proper treatment called antiretroviral therapy (ART).

TARGET POPULATIONS

KEY POPULATION

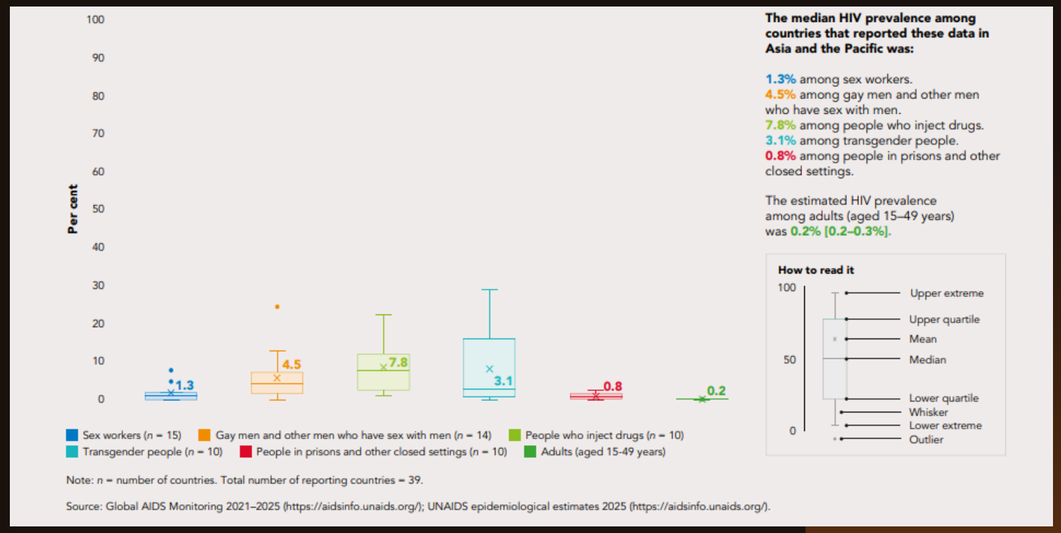
Transgender

Gay men

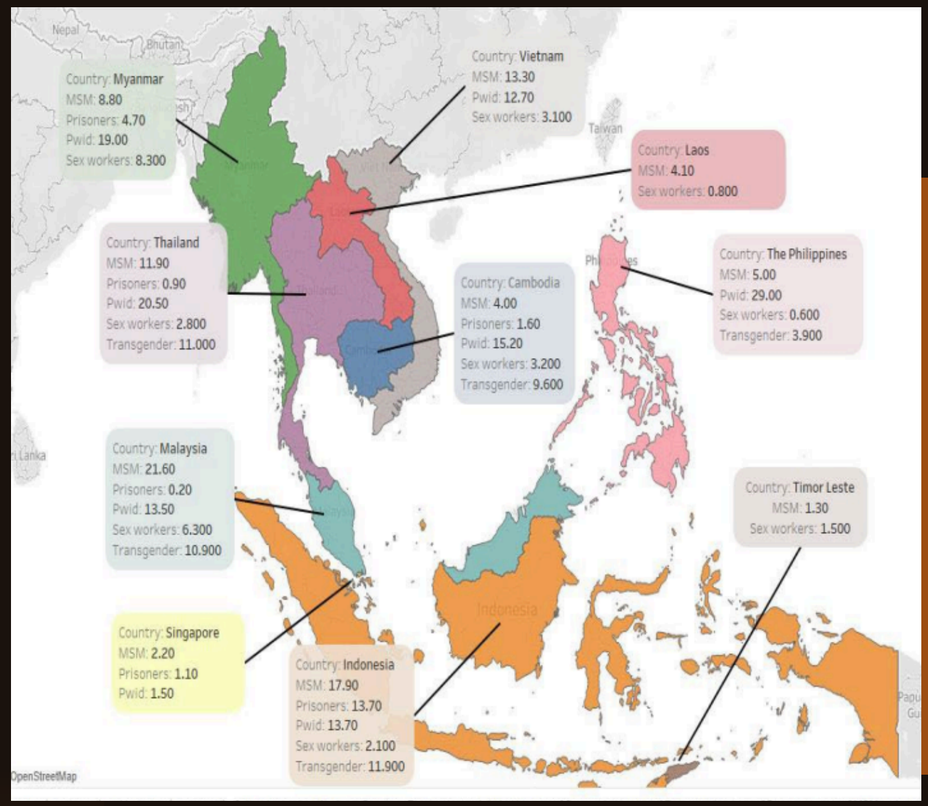
Sex workers

People in prisons and other closed settings

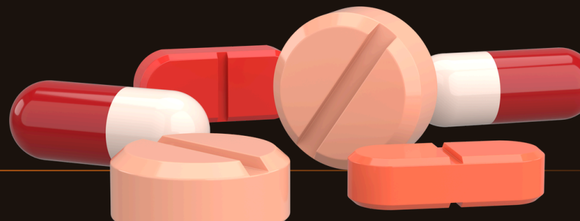
People who inject drugs



SOURCE : UNAIDS 2025 UPDATE



SOURCE : REPORT FROM NATIONAL UNIVERSITY OF SINGAPORE ON UNAIDS 2020 DATA



CRIMES/CASES RELATED TO INFECTION OF HIV/AIDS

CRIME STATISTICS IN MALAYSIA

ACTS INVOLVING CONTROLLED DRUGS OR OTHER PSYCHOACTIVE SUBSTANCES

Number of arrested individuals with positive urine tests 156,803 persons

INJURIOUS ACTS OF A SEXUAL NATURE

Sexual crime 7,740 cases

Number of convicted prisoners admission

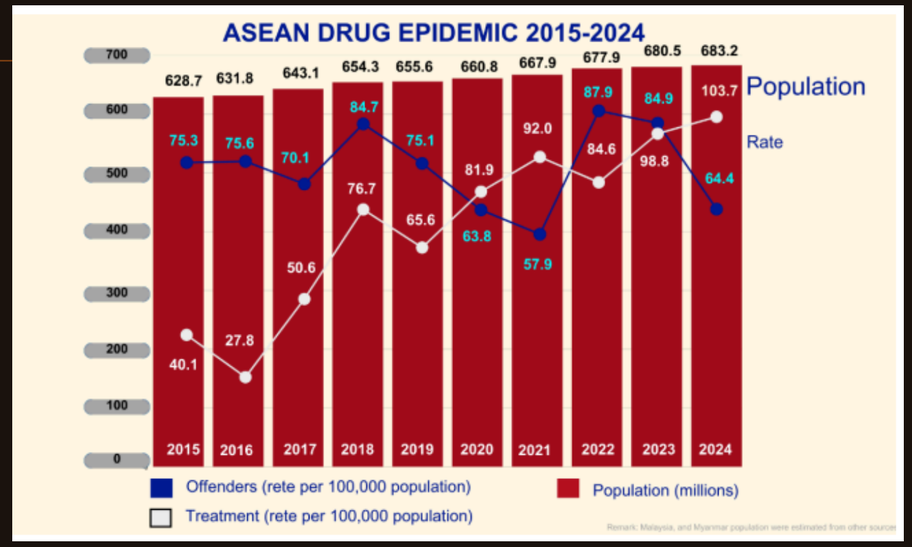
Male 117,211 persons

Female 11,705 persons

Source: Crime Statistics, Malaysia, 2025, Department of Statistics Malaysia (DOSM)

ASEAN CHALLENGES

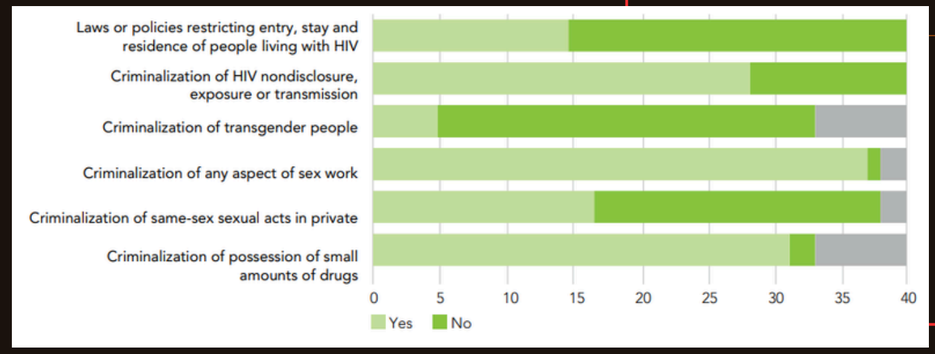
Population and Drug Epidemic in ASEAN



Source: ASEAN Narcotics Cooperation Center Reported on 2025

Number of countries with HIV-related discriminatory and punitive laws, Asia and the Pacific, 2024

SOURCE : UNAIDS 2025 UPDATE



ANTIRETROVIRAL THERAPY

Antiretroviral therapy (ART) is a combination of medications used to treat HIV (human immunodeficiency virus). It works by stopping HIV from reproducing. It can reduce your levels of HIV and keep your immune system healthy. It's not a cure, but many people reach undetectable levels of HIV.

Which drugs are used in antiretroviral therapy?

There are many different types of antiretroviral medications. Each one uses a different strategy, at different points in HIV replication, to stop HIV from making more copies of itself. Types of ART medications include:

- Entry inhibitors.
- Capsid inhibitors.
- Nucleoside reverse transcriptase inhibitors (NRTIs).
- Non-nucleoside reverse transcriptase inhibitors (NNRTIs).
- Integrase inhibitors/integrase strand transfer inhibitors (INSTIs).
- Protease inhibitors.
- Pharmacokinetic enhancers.
- Combination medications.

BENEFITS OF ANTIRETROVIRAL THERAPY (ART)

- *Keeps your T-cell counts higher and your immune system healthy.*
- *Lowers your risk of having a weakened immune system (immunocompromised) and developing serious or fatal infections.*
- *Lowers the risk you'll develop AIDS.*
- *Lowers your risk of developing certain cancers that are more common among people with HIV (like Kaposi sarcoma, non-Hodgkin lymphoma and others).*
- *Prevents HIV from spreading to others through sex if you have an undetectable viral load.*

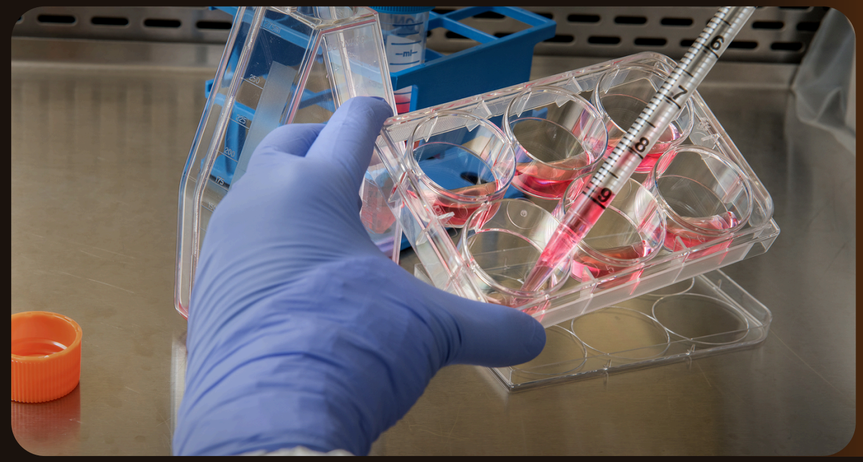
METHODOLOGY

STEP 1

Data Collection

STEP 2

Data Cleaning



STEP 3

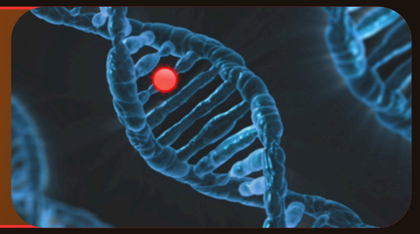
Exploratory Data Analysis (EDA)

STEP 4

Statistical Analysis

STEP 5

Inequality Analysis



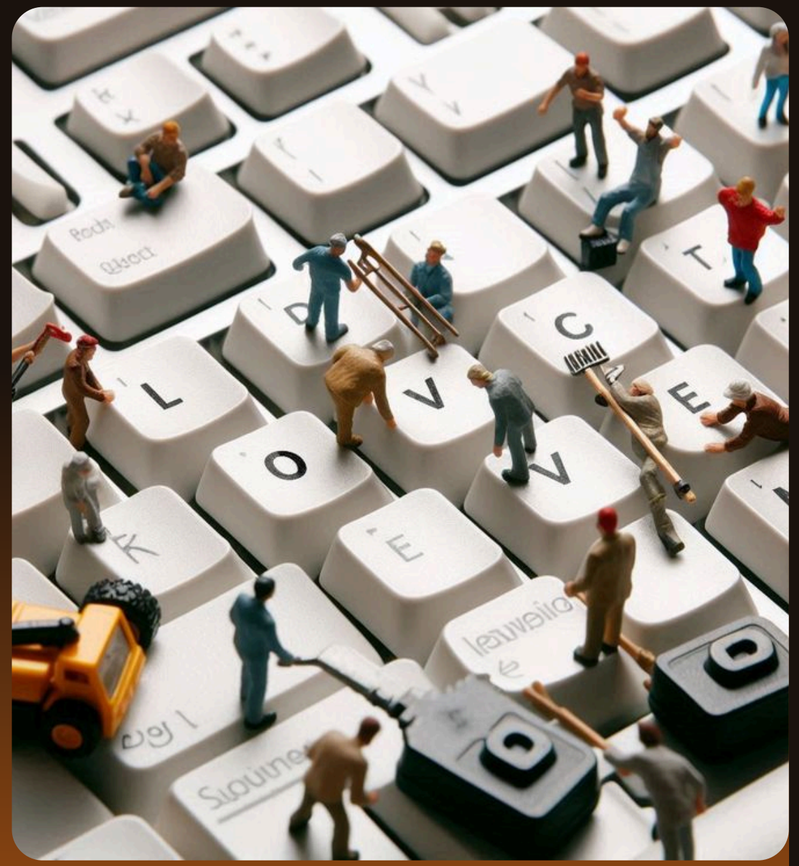
STEP 1 - DATA COLLECTION

The dataset was collected from trusted international health organizations such as:

- World Health Organization
- World Bank
- UNAIDS

STEP 2- DATA CLEANING

Data preprocessing was performed using Python to ensure data quality and consistency.



The cleaning process included:

- ***converting variables from text format to numeric format***
- ***replacing "No data" values with missing values***
- ***extracting numerical estimates from range values***
- ***handling missing values using median imputation***
- ***verifying data types and removing inconsistencies***

The cleaned dataset was then used for further analysis and visualization.

These steps ensure the reliability and accuracy of the analysis results.

STEP 3 - EXPLORATORY DATA ANALYSIS (EDA)

Exploratory data analysis was conducted using Microsoft Power BI to understand the distribution of HIV burden and ART coverage across countries.

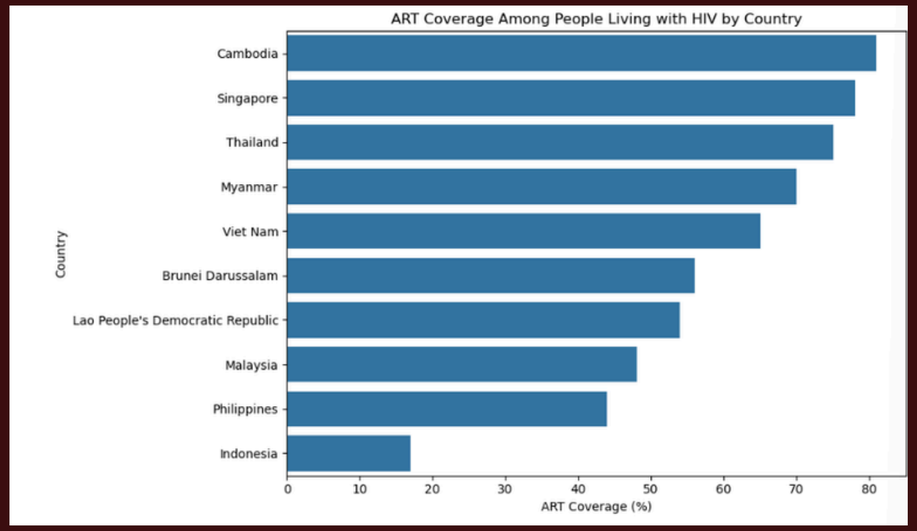
Key visualizations include:

- ***HIV burden by country (estimated number of people living with HIV)***
- ***ART coverage by country***
- ***Comparison of ART coverage and HIV burden across ASEAN countries***

This step helps identify patterns, trends, and disparities in HIV/AIDS distribution within the region.

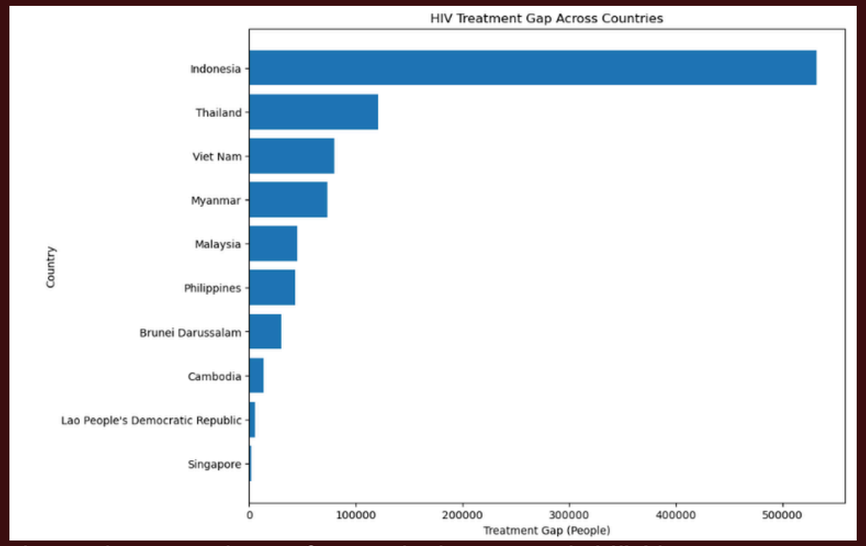
STEP 4 - INEQUALITY ANALYSIS

ART coverage (%)



- Cambodia , Singapore and Thailand have the highest coverage (above 75%) , indicating strong treatment access.
- In contrast ,"Indonesia stands out as an outlier with the lowest ART coverage and highest treatment gap."
- Other countries fall in moderate range (40-70%) .
- Overall , the graph shows that there is clear inequality in ART access across ASEAN countries

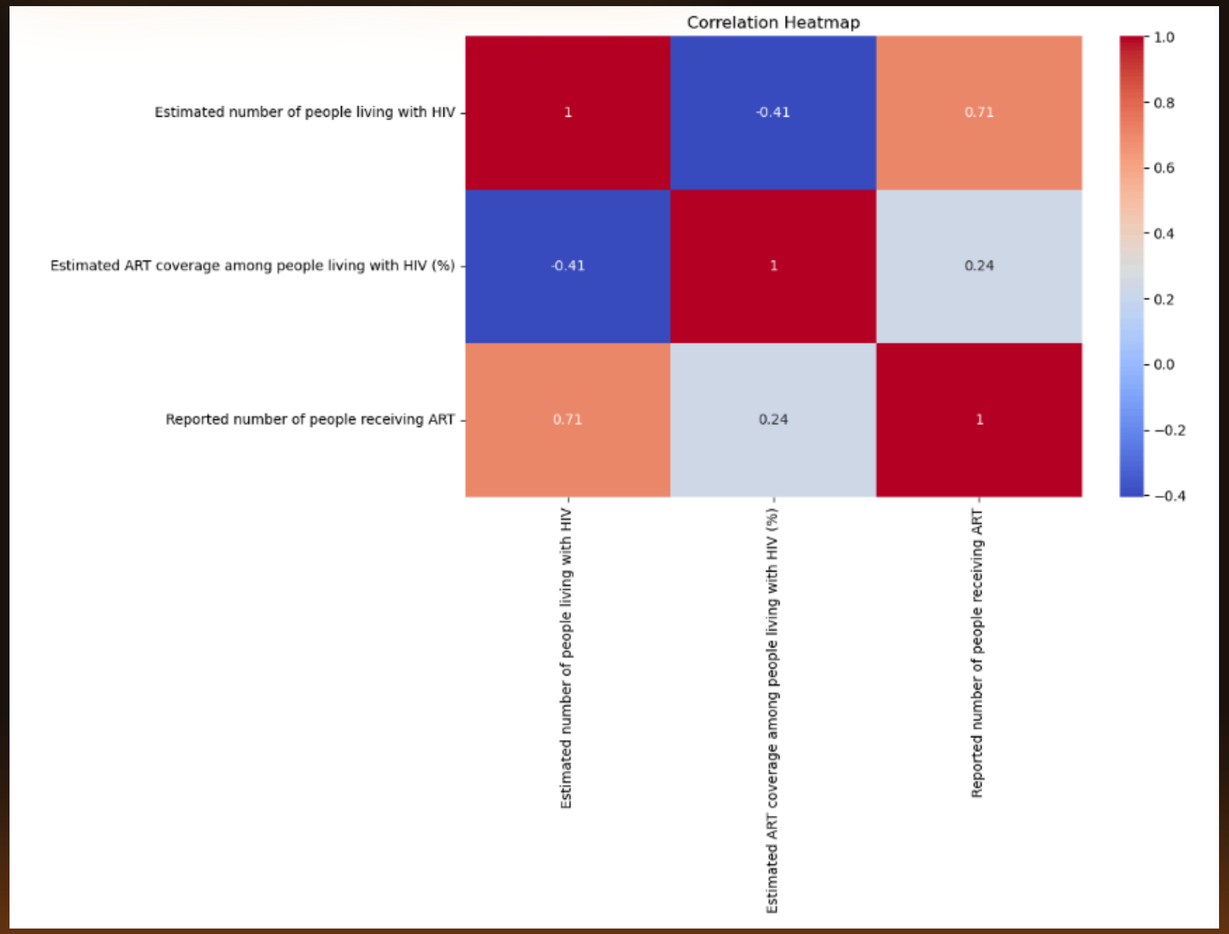
Treatment Gap

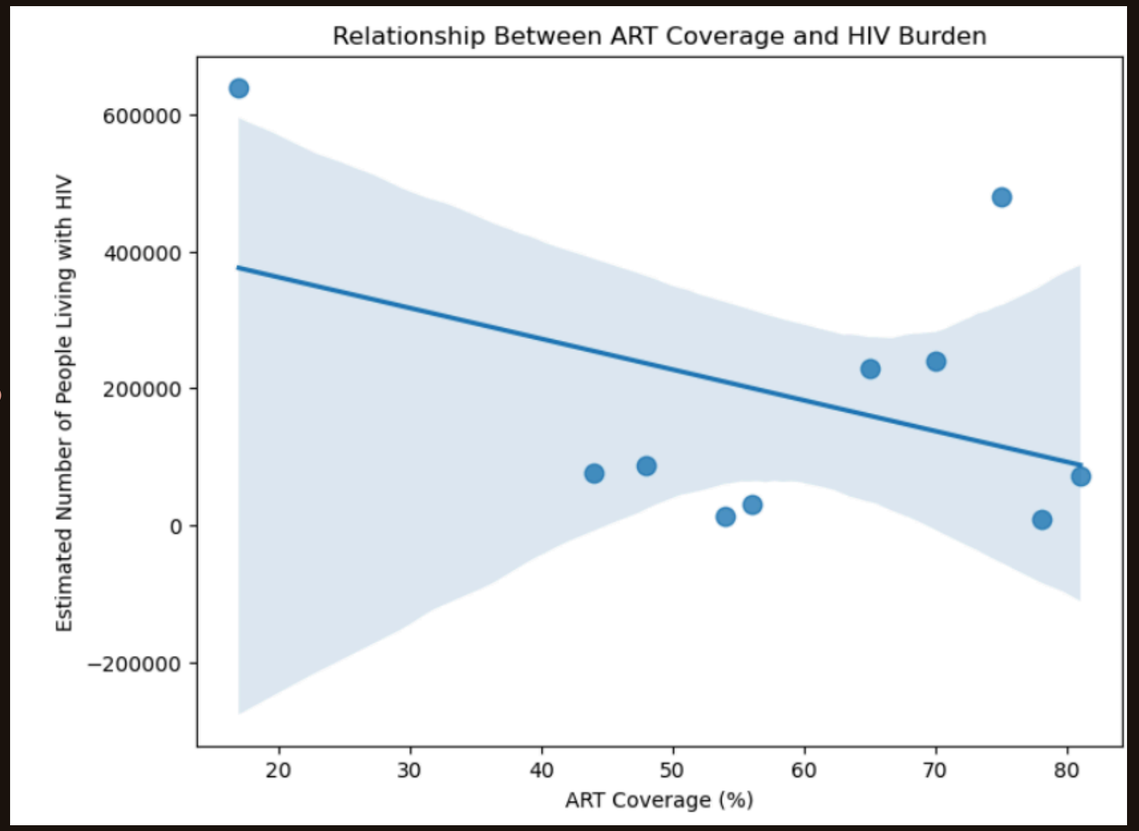


- This graph explain number of people living with HIV but not receiving ART.
- Indonesia has the largest gap by a wide margin , indicating lack of treatment access.
- Countries like Thailand , Viet Nam and Myanmar show moderate gap , while Singapore , Lao PDR and Cambodia , have very small gap , suggest better treatment coverage .
- Overall , this graph highlight clear inequality in access to HIV treatment across countries.

This analysis highlights clear inequality in ART access across countries. Nations with lower ART coverage , tend to have significantly larger treatment gap , indicate limited healthcare access . This pattern is consistent with the negative relationship observed between ART coverage and HIV burden.

RELATIONSHIP BETWEEN ART COVERAGE AND HIV BURDEN





- Widely dispersed data points → variability across countries.
- Moderate negative correlation (r = -0.41)
- Higher ART coverage → lower HIV burden
- Strong positive correlation (r = 0.71)
- Higher HIV burden → more ART provided

Interpretation

- ART access plays a critical role in reducing HIV burden
- Treatment distribution responds to disease prevalence

DATA FINDINGS

- ART coverage varies across countries , indicating inequality
- Higher ART coverage → smaller treatment gap
- Indonesia is an outlier with the largest treatment gap
- Moderate negative correlation between ART coverage and HIV burden(r = -0.41)
- Strong positive correlation between HIV burden and number receiving ART (r = 0.71)

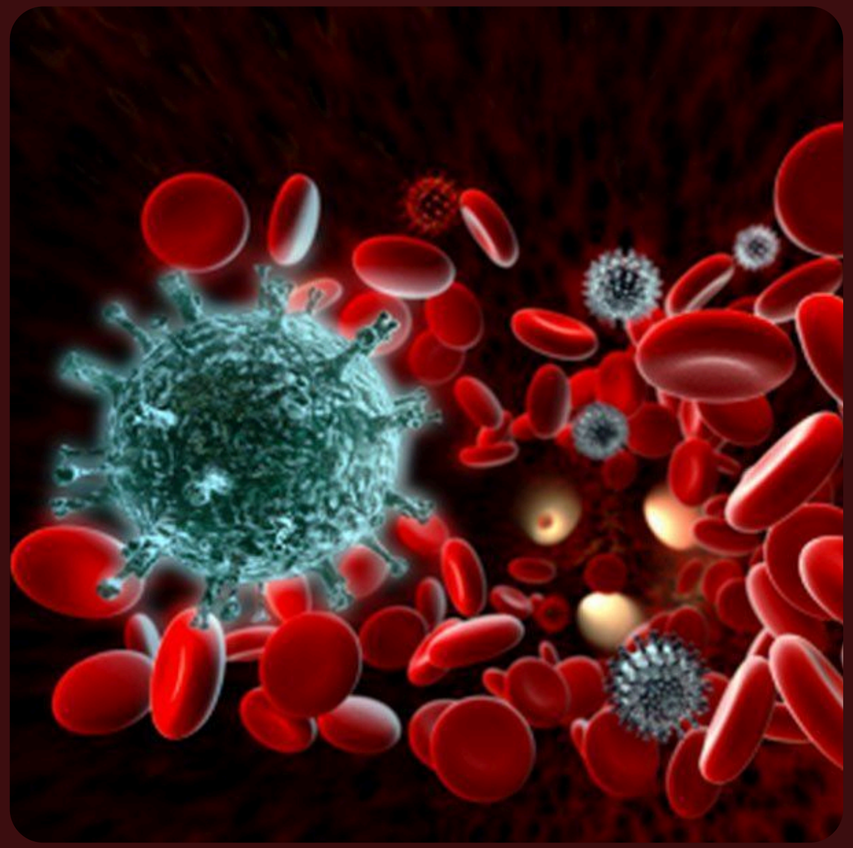
DATA DRIVEN STRATEGIES

Based on the analysis , several data-driven strategies can be proposed by :

- Increase ART accessibility in low-coverage countries to reduce treatment gaps
- Prioritize high-gap countries such as Indonesia for targeted interventions
- Improve early detection through testing programs
- Strengthen healthcare funding and infrastructure
- Enhance data collection for better decision-making

CONCLUSION

- *Clear inequality in ART access across ASEAN countries*
- *Disparities in healthcare access, data, and social factors remain challenges*
- *Inclusive and accurate data is essential for effective intervention*
- *Strengthening data systems and regional cooperation can improve outcomes*
- *Data-driven decisions are key to reducing HIV/AIDS inequality*



BETTER DATA → BETTER DECISIONS → BETTER HEALTH OUTCOMES

THANK YOU